HokieStart - Freshman Onboarding App

GitHub Link: https://github.com/adityasunke/HokieStart

Academic Resource Hub

Use case UC1: Search Academic Resources
Primary Actor: Student
Stakeholders & Interests: Student wants to quickly find tools like Canvas, HokieSpa, ALEKS; Admin: wants search results to reflect curated content
Preconditions: User is logged in
Success Guarantee: Relevant resources are returned and displayed within 1 second
Main Success Scenario: Student navigates to Academic Hub. Student types a keyword into the search bar. System queries resource database, Matching resources are displayed with title, description, and links
Extensions: No results found, system suggests similar resources or prompts admin to add content
Special Requirements: Search must return results in under 1 second
Technology & Data Variation: Search may use keyword matching or fuzzy search

Use case UC2: Bookmark a Resource
Primary Actor: Student
Stakeholders & Interests: Student wants quick access to frequently used resources
Preconditions: User is logged in; resource exists in the system
Success Guarantee: Resource is saved to user's bookmarks and persists across sessions
Main Success Scenario: Student finds a resource. Student taps the bookmark icon, and the system saves the resource to their bookmarks list. Resource appears in the "Bookmarked" section on the hub
Extensions: Resource already bookmarked, tap again to remove it
Special Requirements: None
Technology & Data Variation: Bookmarks stored in user profile in database

Use Case UC3: Admin Manages Resource Library
Primary Actor: System Admin
Stakeholders & Interests: Admin needs to keep resources accurate and up to date; Student expects reliable links and descriptions
Preconditions: Admin is logged in with admin role
Success Guarantee: Resource is added, edited, or removed and changes are immediately visible to students

Main Success Scenario: Admin navigates to admin dashboard. Admin selects "Manage Resources". Admin adds a new resource with name, URL, category, and description. System validates and publishes the resource. Resource appears in student-facing hub
Extensions: URL is invalid, system prompts admin to correct it
Special Requirements: Only admins may add, edit, or delete resources
Technology & Data Variation: None

Use Case UC4: Set Notification Preferences
Primary Actor: Student
Stakeholders & Interests: Student wants to control what notifications they receive; Admin wants important announcements to reach students
Preconditions: User is logged in
Success Guarantee: Notification preferences are saved; student only receives notifications they opted into
Main Success Scenario: Student goes to notification settings in Academic Hub Student toggles on/off notification categories (e.g., deadlines, announcements, new resources) System saves preferences. Future notifications are filtered accordingly
Extensions: Connection failure, system retries save and shows error if persistent
Special Requirements: None
Technology & Data Variation: Push notifications via mobile; email fallback available

Campus Navigation

UC1 - View Campus Map
Primary Actor: Freshman Students
Stakeholders and Interests: Students want an easy way to view the campus layout and nearby locations. The university wants students to navigate campus efficiently.
Preconditions:
- User has opened the onboarding app.
- User is connected to the internet.

Success Guarantee:
The campus map loads successfully and displays buildings and locations.
Main Success Scenario:
- User opens the map feature in the app.
- The campus map is loaded.
- Buildings, landmarks, and important locations appear.
- The user can zoom and move around the map.

Extensions:
If the map fails to load, the system displays an error message.

Special Requirements:
The map should load within 2 seconds.
Technology and Data Variations:
Map data may come from Google Maps or another mapping API.

UC2 - Search for a Location
Primary Actor: Freshman Student
Stakeholders and Interests:
Students want to quickly find buildings, dorms, dining halls, or locations in town.
Preconditions:
User is in the map feature.
Success Guarantee:
The searched location appears on the map.
Main Success Scenario:

- User types a location in the search bar.
- System searches the location database.
- Matching locations are displayed.
- User selects one.
- The map centers on the selected location.

Extensions:
If the location is not found, the system displays "No results found."
Special Requirements:
Search results should appear instantly.
Technology and Data Variations:
Location data may include campus buildings and nearby town locations.

UC3 - Save a Location
Primary Actor: Freshman Student
Stakeholders and Interests:
Students want to save frequently visited places (dorm, library, dining hall).
Preconditions:
User has selected a location on the map.
Success Guarantee:
The selected location is saved to the user's saved places list.
Main Success Scenario:

- User selects a location on the map.
- User taps the "Save Location" button.
- The system stores the location in the user profile.
- The location appears in the saved locations list.

Extensions:

If the user is offline, the location is saved locally.
Special Requirements:
Saved locations must persist across sessions.
Technology and Data Variations:
Locations stored in database or local storage.

UC4 - Generate Route to a Destination
Primary Actor: Freshman Student
Stakeholders and Interests:
Students want directions between campus buildings or town locations.
Preconditions:
- The user has selected a destination.
- Success Guarantee:
- A route is displayed on the map.

Main Success Scenario:
- The user selects a destination.
- The user selects "Get Directions."
- The system calculates the route.
- The route appears on the map.
- The user follows the route.

Extensions:
If no route is available, the system notifies the user.
Special Requirements:
Route calculation should take less than 3 seconds.
Technology and Data Variations:
Routes may be walking routes or driving routes.

Account Setup and User Management

Use Case UC1: Create Account
Primary Actor: Freshman Student
Stakeholders and Interests: Student: wants a quick, simple registration process; Admin: wants verified VT students only in the system
Preconditions: User has a valid Virginia Tech email address (@vt.edu)
Success Guarantee: A new user account is created, stored in the database, and the student is redirected to the onboarding flow
Main Success Scenario:
- Student opens the app and selects "Create Account"
- Student enters their VT email, name, and a password
- System validates the email domain (@vt.edu)

- System sends a verification email to the provided address
- Student clicks the verification link
- Account is created and student is logged in

Extensions: If the email domain is not @vt.edu, the system displays an error and blocks registration. If the verification link expires, the student may request a new one.
Special Requirements: Account creation must complete within 3 seconds of email verification
Technology and Data Variation: Email verification; user credentials stored with hashed passwords.

Use Case UC2: Log In
Primary Actor: Freshman Student
Stakeholders and Interests: Students want fast, secure access to their account; Admins want unauthorized access prevented
Preconditions: Student has an existing verified account
Success Guarantee: Student is authenticated and redirected to the home dashboard
Main Success Scenario:

- Student opens the app and selects "Log In"
- Student enters their email and password
- System validates credentials against the database
- System generates a session token
- Student is redirected to the home dashboard

Extensions: If credentials are wrong, the system displays "Invalid email or password." After 5 failed attempts, the account is temporarily locked for 15 minutes.
Special Requirements: Login response must occur within 2 seconds.
Technology and Data Variation: Session management via JWT tokens; supports OAuth login with Virginia Tech SSO

Use Case UC3: Edit Profile
Primary Actor: Freshman Student
Stakeholders and Interests: Students want to keep personal info accurate and up to date; Admins want profile data to remain valid and consistent
Preconditions: Student is logged in
Success Guarantee: Updated profile information is saved and immediately reflected in the app
Main Success Scenario:

- Student navigates to "My Profile"
- Student selects "Edit Profile"

- Student updates fields such as display name, major, graduation year, or profile photo
- Student taps "Save"
- System validates the inputs and saves changes to the database
- Updated profile is displayed immediately

Extensions: If a required field is left blank, the system prompts the student to fill it in. If the profile photo exceeds the size limit, the system asks the student to upload a smaller file.
Special Requirements: Profile photo uploads must not exceed 5 MB
Technology and Data Variation: Images stored in cloud storage (Firebase Storage) and text fields stored in relational database

Use Case UC4: Reset Password
Primary Actor: Freshman Student
Stakeholders and Interests: Students want to regain account access if the password is forgotten. Admins want the process to be secure and not exploitable
Preconditions: Student has a registered account with a verified email
Success Guarantee: Student successfully resets their password and can log in with the new credentials
Main Success Scenario:

- Student selects "Forgot Password" on the login screen
- Student enters their registered VT email
- System sends a password reset link to that email
- Student clicks the link and is taken to the reset page
- Student enters and confirms a new password
- System updates the password in the database and invalidates the reset link
- Student is redirected to the login screen

Extensions: If the email is not associated with an account, the system displays a generic message ("If this email is registered, a reset link has been sent") to prevent account enumeration. If the reset link has expired, the student may request a new one.
Special Requirements: Reset links must expire after 30 minutes; new password must meet minimum complexity requirements (8+ characters, at least one number)
Technology and Data Variation: Reset tokens stored with expiration timestamps; email delivery via SMTP

Onboarding and Task Management

Use Case UC1: Complete Onboarding Checklist

Actor Type: Student.
Stakeholders and Interests: The student would like to receive a guided introduction of the Virginia Tech resources; the university would like to see all freshmen complete key setup steps.
Preconditions: The user must create an account and log into the system for the first time. **Success Guarantee**: Upon successful completion of all onboarding steps, the student is marked as onboarded in the system.
Main Success Scenario: After logging in for the first time, the student will have a checklist presented to them for onboarding. As the student finishes each checklist item (e.g., setting up email, reviewing campus maps, joining a community), the system will automatically mark it as completed. Upon finishing all checklist items, the student will receive a 'congratulations' message as well as access to the entire app dashboard.
Extensions: The student will have the choice to skip a checklist item; however, the system will save that portion of the student's progress until they return to complete it.
Special Requirements: The current status of checklist completion will carry forward from session to session.
Technology/Data Variations: The current status of the checklist will be saved in the user profile database.

Use Case UC2: Create a Personal Task
Primary Actor: Student
Stakeholders & Interests: Students want to track personal to-dos like assignments and appointments.
Preconditions: User is logged in.
Success Guarantee: Task is saved and visible in the student's task list.
Main Success Scenario: Student navigates to the Task Manager. The student taps "Add Task" and enters a title, due date, and optional description. System saves the task. Task appears in the student's task list sorted by due date.
Extensions: Student leaves required fields blank; system prompts them to fill in a title before saving.
Special Requirements: None.
Technology & Data Variation: Tasks stored in user profile database.

Use Case UC3: Mark a Task as Complete
Primary Actor: Student
Stakeholders & Interests: Student wants to track progress and feel a sense of accomplishment.
Preconditions: User is logged in; at least one task exists in their list.
Success Guarantee: Task is marked complete and moved to a completed section.
Main Success Scenario: Student views their task list. Student taps the checkbox next

to a task. System marks the task as complete. Task moves to the "Completed" section of the task list.
Extensions: Student accidentally marks a task complete; student can tap again to unmark it.
Special Requirements: None.
Technology & Data Variation: None.

Use Case UC4: Receive Task Deadline Reminder
Primary Actor: Student
Stakeholders & Interests: Student wants to be reminded of upcoming deadlines so nothing is missed.
Preconditions: User is logged in; student has at least one task with a due date; notifications are enabled.
Success Guarantee: Student receives a reminder notification before the task deadline.
Main Success Scenario: System detects a task due within 24 hours. System sends a push notification to the student's device with the task title and due time. Student taps the notification and is taken directly to the task.
Extensions: Notifications are disabled; system displays an in-app banner reminder instead.
Special Requirements: Reminder must be delivered at least 24 hours before the deadline. Technology & Data Variation: Push notifications on mobile; email fallback available.

Community and Support

Use Case UC1: Post a Discussion Forum Question
Primary Actor: Student
Stakeholders & Interests: Student wants to ask questions and get helpful answers from peers or staff; Admin wants the forum to stay organized and free of inappropriate content
Preconditions: User is logged in with a valid VT email account
Success Guarantee: A new forum post is created, categorized correctly, and visible to all users; the author is notified of any replies
Main Success Scenario: Student navigates to the Discussion Forum section. Student selects a category (e.g., Academic, Campus Life, Technical Help). Student enters a title and body text for the post. Student optionally attaches a tag (e.g., course number). Student submits the post. System validates the content and publishes the post. System displays a confirmation message.

Extensions: Student leaves title or body empty, system displays an error and prompts them to complete the required fields. Content contains prohibited language, system flags the post and holds it for moderation review before publishing.
Special Requirements: Content must be filtered for inappropriate or harmful language before publishing
Technology & Data Variation: Student may submit the post via a mobile device with a responsive interface

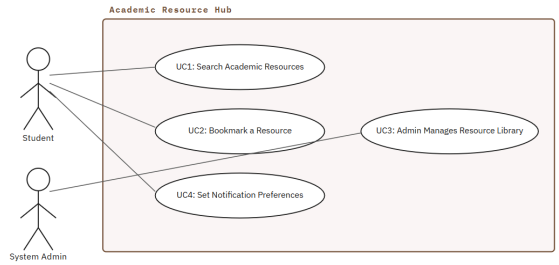
Use Case UC2: Create and Join a Study Group
Primary Actor: Student
Stakeholders & Interests: Student (creator) wants to form a study group for a specific course; Student (joiner) wants to find and join a relevant group; Admin wants study groups to be organized and appropriate
Preconditions: User is logged in with a valid VT email account
Success Guarantee: A new study group is created and listed for the selected course, or the student is successfully added to an existing study group
Main Success Scenario: Student navigates to the Study Groups section. Student browses existing groups or searches by course name/number. To create: student selects "Create New Study Group," enters a group name, selects the course, sets a max member count, and submits. System creates the group and adds the student as owner. To join: student selects an existing group and clicks "Join Group." System adds the student to the group. System displays a confirmation message.
Extensions: Group name already exists for the same course, system suggests joining the existing group or choosing a different name. Study group is full, system notifies the student and suggests creating a new one.
Special Requirements: None
Technology & Data Variation: None

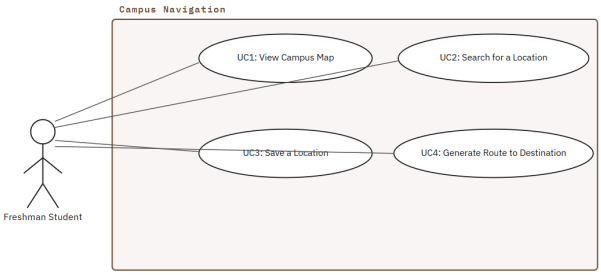
Use Case UC3: Search the FAQ
Primary Actor: Student
Stakeholders & Interests: Student wants to quickly find answers to common questions without waiting for a response; Admin wants to maintain an up-to-date FAQ to reduce repetitive forum posts
Preconditions: User is logged in; FAQ database has been populated by a System Admin
Success Guarantee: Student receives relevant FAQ results matching their search query; if no results match, the student is offered an option to post a new forum question
Main Success Scenario: Student navigates to the FAQ section. Student enters a keyword or question in the search bar. System searches the FAQ database and returns matching results ranked by relevance. Student selects a FAQ entry to view the full answer. System displays the full FAQ answer.

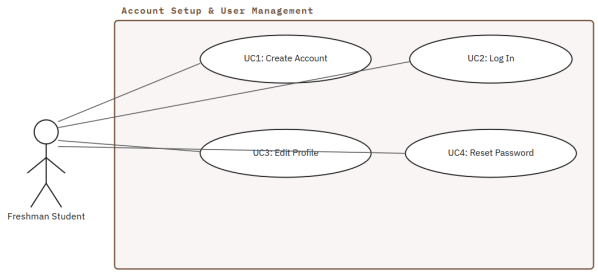
Extensions: No FAQ entries match the search query, system displays "No results found" and suggests posting a new question on the Discussion Forum. Student finds the answer unhelpful, student clicks "Not Helpful" and the system logs the feedback for admin review.
Special Requirements: Search results should be returned within 2 seconds
Technology & Data Variation: Student may use voice-to-text on mobile to enter the search query

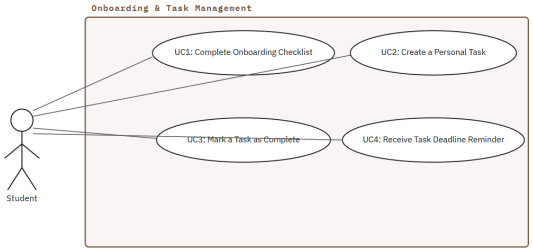
Use Case UC4: Submit Feedback or Report an Issue
Primary Actor: Student
Stakeholders & Interests: Student wants to report a bug, suggest a feature, or provide general feedback; Admin wants to receive organized, categorized feedback to prioritize improvements
Preconditions: User is logged in with a valid VT email account
Success Guarantee: Feedback is submitted, categorized, and stored; student receives a confirmation with a tracking reference number; admin can view and manage the submission
Main Success Scenario: Student navigates to the Feedback & Support section. Student selects a category (Bug Report, Feature Request, General Feedback). Student enters a subject line and detailed description. Student optionally attaches a screenshot or file. Student submits the form. System validates and stores the submission. System displays a confirmation with a reference number. System notifies the relevant admin.
Extensions: Student leaves required fields empty, system highlights missing fields and prompts completion. Attached file exceeds 10 MB limit, system displays an error and prompts the student to reduce file size. System detects a duplicate submission, notifies the student and asks for confirmation before submitting.
Special Requirements: Attachments limited to 10 MB; supported formats: PNG, JPG, PDF
Technology & Data Variation: Student may capture a screenshot directly from the app using a built-in tool

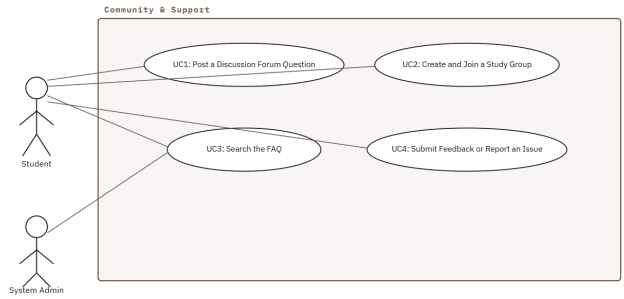
2) Use Case Diagrams:



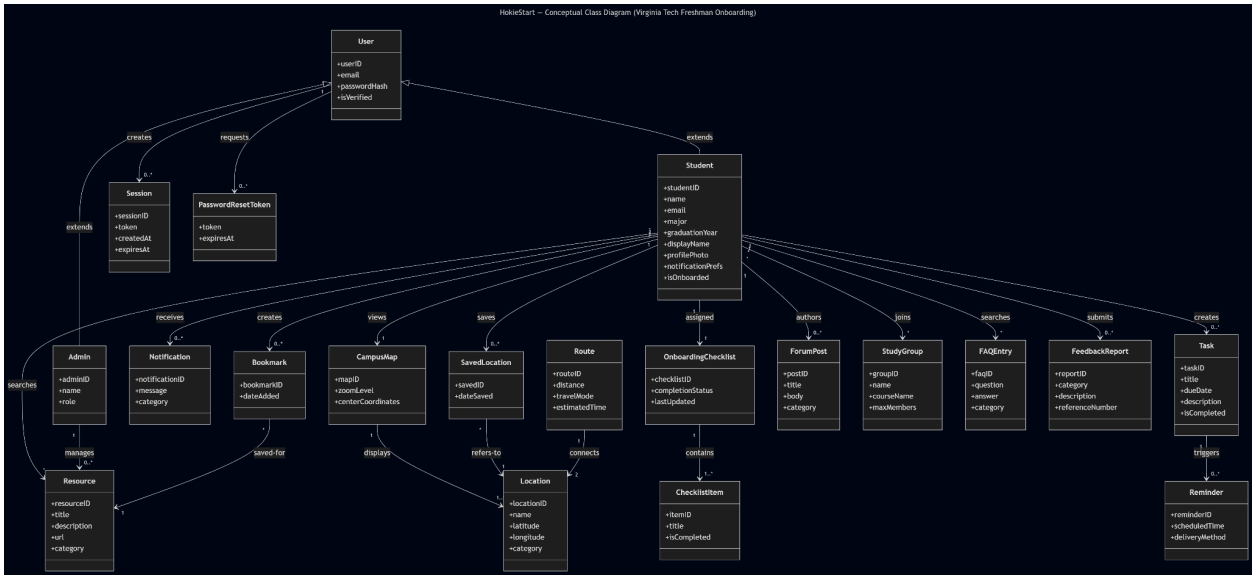








3) Conceptual Class Diagrams:



4) Supplementary Specifications:

Sankalp Dasari: Resource search must return results in under 1 second; admin editing of resources must be restricted via role-based access control; database must maintain consistency with no duplicate resources

Aditya Sunke:
The map should load within 2 seconds under normal wifi conditions.

Aryan Shiva: Passwords must always be stored securely using hashing and never saved in plain text, and session tokens should automatically expire after 24 hours to protect user accounts. Login and account creation should respond within 2–3 seconds under normal network conditions to keep the experience smooth. Registration is limited to @vt.edu email addresses, and student data must never be shared with third parties in compliance with FERPA guidelines.

Ishaan Jain: The onboarding checklist must load within 2 seconds. Task reminders must be delivered reliably with no more than a 1-minute delay from the scheduled notification time.

Varun Singh: All forum posts and messages must be transmitted over HTTPS with TLS encryption; the system must support at least 500 concurrent users browsing the forum and FAQ without response time degradation; all user-generated content must pass through automated content filtering for profanity, harassment, and spam before publishing.